

Text Classification in NLP

CMPE346 — Group 2

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Group task: Text Classification

Each member selected a different NLP classification application and a dedicated dataset. All three fine-tuned the same set of models and reported Accuracy and weighted F1.

Student	Task
Atakan Gül	Media Bias Detection in News
[Student 2]	[Topic]
[Student 3]	[Topic]

BiLSTM

Bidirectional LSTM trained from scratch. Captures sequential context in both directions. Serves as the classical deep learning baseline.

BERT

Transformer encoder pretrained via masked language modeling. Fine-tuned end-to-end. `bert-base-uncased`.

RoBERTa

Optimized BERT: removes next sentence prediction, uses dynamic masking and more data. `roberta-base`.

Task: Classify a news sentence as *biased* or *not biased*.

Dataset: BABE (Bias Annotations By Experts)

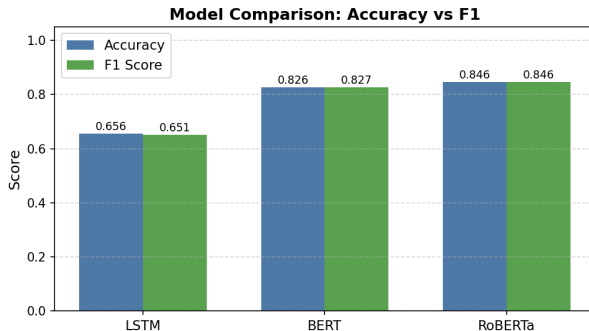
- ▶ 4,121 expert-annotated news sentences
- ▶ Diverse outlets, binary labels
- ▶ 3,121 train / 1,000 test

Bias is expressed through framing and tone — not false facts. This makes it a genuinely hard classification problem.

Model	Acc.	F1
BiLSTM	0.656	0.651
BERT	0.826	0.827
RoBERTa	0.846	0.846

RoBERTa best. +19% over BiLSTM.

Transformers benefit from pretraining on small labeled data.



Task: [Description]

Dataset: [Name — brief stats]

Model	Acc.	F1
BiLSTM	—	—
BERT	—	—
RoBERTa	—	—

[Student 2 key finding]

Task: [Description]

Dataset: [Name — brief stats]

Model	Acc.	F1
BiLSTM	—	—
BERT	—	—
RoBERTa	—	—

[Student 3 key finding]

Conclusion

- ▶ RoBERTa outperforms BERT and BiLSTM across all three tasks
- ▶ Pretraining is decisive — BiLSTM trained from scratch cannot compete on small datasets
- ▶ The performance gap between LSTM and transformers is consistent regardless of the classification domain
- ▶ Future work: domain-adaptive pretraining, larger model variants